

WEEK 2 - EXPERIMENT 2

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Explore CSS3 Styling to Design a Responsive Student Learning Portal Webpage

2. AIM

To understand how CSS3 enhances the appearance and layout of a webpage by designing a styled "Student Learning Portal" page using flexbox, box-shadow, border-radius and hover effects.

3. OBJECTIVE

- To understand the basic syntax and structure of CSS3.
- To apply CSS selectors, colors, fonts, borders, margins and padding.
- To style a webpage using background colors, box-shadow and rounded corners (border-radius).
- To arrange content cards responsively using CSS3 Flexbox (display: flex, gap, flex-wrap).
- To implement hover effects for navigation links and buttons.
- To structure a complete webpage with a header, navigation bar, welcome section, content cards and footer.

4. THEORY

CSS3 (Cascading Style Sheets, Level 3) is the latest evolution of the CSS language used to control the presentation and layout of HTML documents. It introduces advanced styling features such as box-shadow, rounded corners (border-radius) and hover-based transition effects, allowing developers to design visually appealing web pages without relying on images or scripts.

CSS rules follow the syntax selector { property: value; }. Selectors can target elements by tag name, class (.class), id (#id), or pseudo-classes such as :hover, which applies a style only while the mouse pointer is over an element.

CSS3 Flexbox is a one-dimensional layout model that makes it easy to align and distribute space among items in a container, even when their size is unknown. Setting display: flex on a container (such as the course-listing section) and using properties like justify-content, gap and flex-wrap allows child elements (the course cards) to be laid out in a row, evenly spaced, and to wrap gracefully on smaller screens.

Together, these CSS3 features allow a single webpage to combine a styled header, navigation bar, an introductory welcome panel and a responsive set of content cards with a consistent visual theme, demonstrating declarative styling and responsive layout design.

5. ALGORITHM / PROCEDURE

- Create a new HTML file and set up the basic HTML5 document structure.

- Add a <style> block and define general/reset styles (margin, padding, box-sizing).
- Style the body with a font family and a light background color.
- Create a header section titled "Student Learning Portal" with background color, text color and a subtitle.
- Create a navigation bar with Home, Courses, About and Contact links; style the links and add a :hover effect with a color change.
- Create a welcome section with a heading and description, styled with padding, border-radius and box-shadow.
- Create a courses section using display: flex to arrange multiple course cards in a row with even spacing.
- Create three course cards (HTML5, CSS3, JavaScript), each with a heading, description and a "Learn More" button, styled with border-radius and box-shadow.
- Add a :hover effect on the buttons to change their background color.
- Create a footer with background color and centered copyright text.
- Open the HTML file in a browser and verify the layout and styling.
- Test hover effects on navigation links and buttons, and verify the responsive card layout renders correctly.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Learning Portal</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: Arial, sans-serif;
      background-color: #f1f5f9;
      color: #333;
    }
    /* Header */
    header {
      background-color: #2563eb;
      color: white;
      padding: 25px;
      text-align: center;
    }
    /* Navigation */
    nav {
      background-color: #1e40af;
      padding: 15px;
      text-align: center;
    }
    nav a {
      color: white;
      text-decoration: none;
      margin: 20px;
      font-size: 16px;
    }
    nav a:hover {
      color: yellow;
    }
    /* Main Section */
    main {
      padding: 40px;
    }
    .welcome {
      background-color: white;
      padding: 30px;
      text-align: center;
```

```

border-radius: 10px;
margin-bottom: 30px;
box-shadow: 0px 4px 10px #ccc;
}
.welcome h2 {
color: #2563eb;
margin-bottom: 15px;
}
.courses {
display: flex;
justify-content: center;
gap: 25px;
flex-wrap: wrap;
}
/* Course Cards */
.course-card {
width: 250px;
background-color: white;
padding: 25px;
text-align: center;
border-radius: 10px;
box-shadow: 0px 4px 10px #ccc;
}
.course-card h3 {
color: #2563eb;
margin-bottom: 15px;
}
.course-card p {
line-height: 1.5;
margin-bottom: 15px;
}
.course-card button {
background-color: #2563eb;
color: white;
border: none;
padding: 10px 18px;
border-radius: 5px;
cursor: pointer;
}
.course-card button:hover {
background-color: #1e40af;
}
/* Footer */
footer {
background-color: #1e293b;
color: white;
text-align: center;
padding: 20px;
margin-top: 40px;
}
</style>
</head>
<body>
<header>
<h1>Student Learning Portal</h1>
<p>Learn &bull; Practice &bull; Grow</p>
</header>
<nav>
<a href="#">Home</a>
<a href="#">Courses</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>
<main>
<section class="welcome">
<h2>Welcome to Our Learning Platform</h2>
<p>
Explore different technologies and improve your
programming and web development skills.
</p>
</section>
<section class="courses">
<article class="course-card">

```

```

<h3>HTML5</h3>
<p>
Learn how to create the structure of modern webpages.
</p>
<button>Learn More</button>
</article>
<article class="course-card">
<h3>CSS3</h3>
<p>
Learn how to design attractive and responsive webpages.
</p>
<button>Learn More</button>
</article>
<article class="course-card">
<h3>JavaScript</h3>
<p>
Learn how to make webpages interactive and dynamic.
</p>
<button>Learn More</button>
</article>
</section>
</main>
<footer>
<p>&copy; 2026 Student Learning Portal | Designed using HTML5 & CSS3</p>
</footer>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
display: flex	Arranges child elements (course cards) in a row and lets them wrap and align evenly using justify-content and gap.
gap	Adds consistent spacing between the flex items (course cards) without using margins on each card.
flex-wrap: wrap	Allows the course cards to move to the next line on smaller screens instead of overflowing.
:hover	Pseudo-class that applies a style only while the mouse pointer is over the element (nav links and buttons).
border-radius	Rounds the corners of an element; used on the welcome section, course cards and buttons.
box-shadow	Adds a soft drop shadow around the welcome section and course cards to create a card-like, elevated look.
box-sizing: border-box	Includes padding and border within an element's declared width/height, keeping layouts predictable.
line-height	Increases the vertical spacing between lines of paragraph text inside each course card for readability.

8. TEST CASES AND EXECUTION DATA

Each styling feature and layout element was verified individually in the browser.

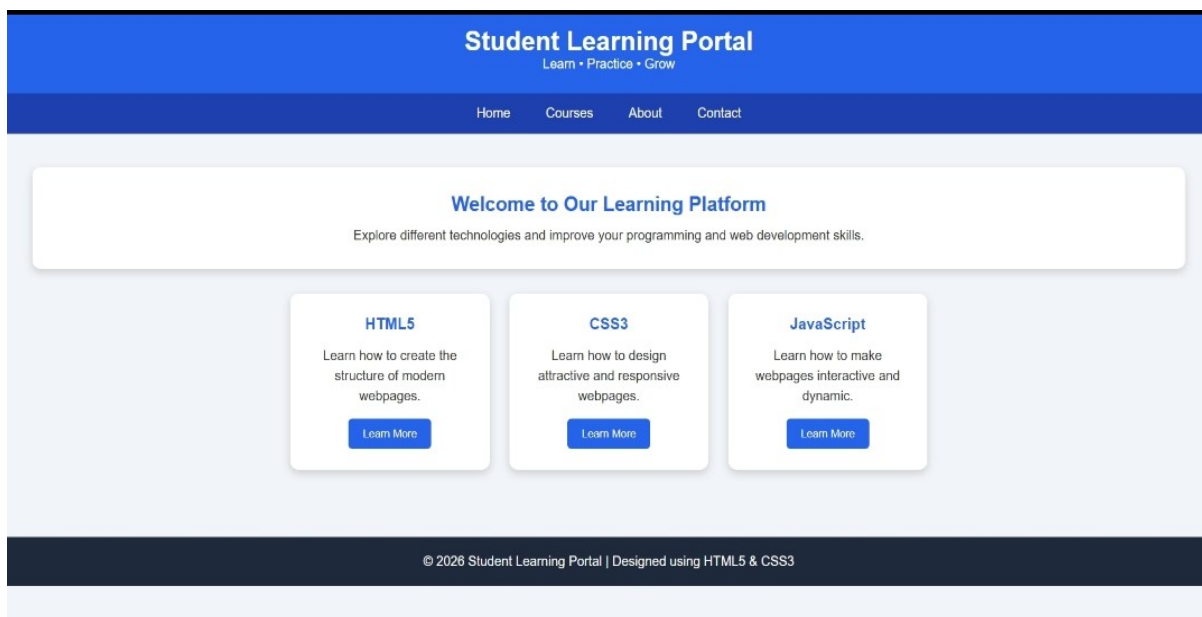
Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-01	Page load	Header, nav, welcome section, course cards and footer render correctly	Rendered correctly	Pass

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-02	Header & nav styling	Header shows blue background with title/subtitle; nav bar shows dark blue background	Displayed correctly	Pass
TC-03	Nav link hover	Link text turns yellow on hover	Hover effect worked	Pass
TC-04	Welcome section styling	White card with rounded corners and box-shadow appears centered	Displayed correctly	Pass
TC-05	Course card layout	Three cards (HTML5, CSS3, JavaScript) align in a row using flexbox with equal spacing	Cards aligned correctly	Pass
TC-06	Card styling	Cards show rounded corners and box-shadow	Displayed correctly	Pass
TC-07	Button hover	"Learn More" button background changes to darker blue on hover	Hover effect worked	Pass
TC-08	Footer	Footer displays dark background with centered copyright text	Displayed correctly	Pass

9. OUTPUT

Output 1: Full Page View

The page displays a styled blue header titled "Student Learning Portal" with the subtitle "Learn • Practice • Grow", a dark blue navigation bar with Home, Courses, About and Contact links (turning yellow on hover), a white welcome panel with rounded corners and a soft shadow, three evenly spaced course cards (HTML5, CSS3 and JavaScript) each with a heading, description and a "Learn More" button, and a dark footer with the copyright text, all rendered correctly in the browser.



10. RESULT

Thus, a webpage titled "Student Learning Portal" was successfully designed using CSS3 styling techniques (box-shadow, border-radius, flexbox layout and hover effects). All components — header, navigation, welcome section, course cards and footer — were verified in the browser and the required visual effects were achieved as expected.